Multimodal Transformer-Based Area Estimation in Remote Sensing

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Abstract — **This study investigates whether incorporating textual descriptions of satellite images improves land cover composition estimation compared to image only approaches, and whether backbone capacity moderates the benefit of multimodal fusion. The task is formulated as an unconstrained multi-output regression problem predicting proportional coverage of seven land cover classes. Three vision transformer backbones namely DeiT, SwinV2, and MaxViT are evaluated in image only and multimodal configurations, with text features encoded by MiniLM, DistilBERT, and RemoteCLIP and fused through a gated mechanism. Results show that multimodal fusion consistently improves performance across all backbones, with MAE reductions of 22.26%, 24.36%, and 18.81% respectively.**

I. Introduction

Accurate land cover composition estimation from remote sensing imagery is a fundamental task in environmental monitoring and agricultural land use mapping. Traditional pixel-level segmentation approaches require dense annotations and significant computational resources, yet many applications require only aggregate class proportions rather than per-pixel labels. Regression based approaches that directly estimate class coverage from image level features offer a more efficient alternative, removing the need for dense annotation.

Recent advances in vision transformers and multimodal learning suggest that attention based architectures complemented by textual information can further improve such estimation tasks. This project investigates this direction by treating land cover estimation as an unconstrained multi-output regression problem, predicting proportional coverage of seven classes from satellite image patches and associated textual descriptions. The study is guided by two research questions: whether incorporating textual descriptions improves land cover proportion estimation over image-only models, and whether backbone capacity moderates the benefit of multimodal fusion. These questions motivate a systematic comparison across three vision transformer backbones, three text encoders, three caption types, and five fusion strength values.

II. Literature

Recent studies highlight both the potential and limitations of current approaches for regression tasks in Earth Observation (EO). A recent perspective on Vision-Language Models emphasizes that, although multimodal models have achieved strong performance in perception tasks, their application to continuous regression problems in EO remains largely unexplored, with challenges such as representation mismatch [1]. At the same time, existing work on fractional land cover estimation demonstrates that area estimation can be effectively modeled as a regression problem directly from satellite imagery, without relying on pixel wise segmentation [2]. Research in remote sensing has increasingly explored multimodal learning by combining image and text representations. For example, transformer based multimodal frameworks, such as those integrating Swin Transformer with language image pretraining, have shown improved performance in tasks like flood image classification [3]. Similarly, vision language models have demonstrated the potential of joint image text representations for understanding complex spatial patterns in EO data [4]. However, existing studies primarily focus on classification or retrieval tasks, with limited attention to regression based problems such as area estimation. This highlights a gap in leveraging multimodal transformer architectures for continuous prediction tasks, which motivates the research question of whether image text fusion can improve land cover proportion estimation.

III. Methodology

A. Dataset

Experiments were conducted on a synthetic remote sensing dataset of 10,000 samples selected through stratified sampling based on dominant land cover class, split into 80% train, 10% validation, and 10% test. Each sample consists of a satellite image paired with textual captions describing the land cover composition, and seven continuous target variables representing proportional coverage of Tree, Shrub, Grass, Crop, Built-up, Barren, and Water classes. Three caption types were evaluated: text-only captions generated by Qwen3-4B, vision-based captions by Vision-Gemma3-4B, and hybrid captions by Gemma3-4B. Prior to training, all captions were preprocessed by removing numerical values, percentages, and leakage prone expressions to ensure performance improvements originated from semantic content rather than explicit target cues.

B. Multimodal Architecture

The proposed framework combines image and text representations for land cover proportion prediction. Three pretrained vision transformer backbones were investigated, namely DeiT-Tiny, SwinV2-Tiny, and MaxViT-Tiny. These are referred to as DeiT, SwinV2, and MaxViT throughout for brevity. DeiT and MaxViT were trained using 224 × 224 image inputs, whereas SwinV2 used 256 × 256 inputs. For all backbones, the original classification head was replaced with a regression head consisting of layer normalization, a fully connected layer with 256 hidden units, ReLU activation, dropout, and a final linear layer producing seven outputs corresponding to the target land cover classes. Textual information was encoded using three frozen text encoders which are MiniLM, DistilBERT, and RemoteCLIP. MiniLM and DistilBERT were selected as lightweight general purpose language models, while RemoteCLIP was included as a domain specific encoder pretrained on remote sensing imagery and text. Keeping the text encoders frozen allowed the study to focus on the effect of multimodal fusion rather than language model fine-tuning. Image and text representations were combined through a gated fusion mechanism. Text features were first projected into the image feature space and then adaptively integrated with visual features through a learnable gate. The projected text features are additionally scaled by a factor α before fusion, controlling the overall contribution of textual information. This design enables the model to determine the contribution of textual information.

C. Experimental Design and Ablation Studies

The experimental design was structured to evaluate the influence of both textual content and text representation on land cover proportion estimation. For each image backbone, all combinations of the three caption types and three text encoders were evaluated under identical training conditions, resulting in nine multimodal configurations per backbone. In addition, an image only baseline was trained for each backbone to quantify the contribution of textual information. This design enables a direct comparison of caption type strategy, text encoder choices, and image backbone capacity. The inclusion of image only baselines allows the contribution of textual information to be measured consistently across architectures, allowing the investigation of backbone capacity effects. Following the comparison of caption types and text encoders, a text scale ablation study was conducted to analyze the influence of textual information in the fusion process. For each image backbone, the best performing caption and text encoder combination was selected based on validation MAE. The text scaling parameter was then varied across five values: 0.3, 0.5, 0.7, 1.0, and 1.5. The objective of this analysis was to determine how strongly textual features should contribute to the final multimodal representation and whether performance improvements were sensitive to fusion strength. Results were evaluated using both overall and class-wise MAE and were compared against the corresponding image only baselines.

D. Training and Evaluation

All models were trained using AdamW with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4}. Training was performed with a batch size of 32 for a maximum of 15 epochs and early stopping with a patience of three. Gradient clipping with a maximum norm of 1.0 was applied throughout training, and all experiments used a fixed random seed of 42. Text inputs were truncated to a maximum length of 128 tokens. For all caption type and encoder experiments, the text scaling factor was fixed at $\alpha = 0.7$. The scaling factor was subsequently varied in the text scale ablation studies. Models were optimized using Mean Absolute Error (MAE) loss. Performance was evaluated using overall MAE and class-wise MAE. All experiments were conducted using identical training settings and data splits to ensure fair comparisons. Training progress and evaluation metrics were tracked using Weights & Biases.

IV. Results

A. Image Only Baseline

Prior to incorporating textual information, each backbone was evaluated in an image only configuration to establish a reference point for measuring the contribution of multimodal fusion. DeiT achieved a baseline MAE of 2.81, SwinV2 achieved 2.48, and MaxViT achieved 2.35. Across all three backbones, Grass and Crop consistently showed the highest class-wise error, while Shrub, Built-up, and Water were predicted most accurately.

B. Ablation Study: Caption Type and Text Encoder

Text only captions consistently outperformed hybrid and vision-based captions across all backbones and encoders. The best text only configurations achieved MAEs of 2.32, 1.88, and 1.92 for DeiT, SwinV2, and MaxViT respectively, representing relative improvements of 17.5%, 24.4%, and 18.3% in MAE over the corresponding image only baselines. Hybrid captions ranked second, with best configurations of MAEs 2.52, 2.08, and 2.14 for DeiT, SwinV2, and MaxViT respectively. Vision based captions performed worst, with best configurations of 2.86, 2.33, and 2.33 MAE respectively. Several vision based configurations on DeiT fell below the image-only baseline, indicating that for weaker backbones, poorly informative captions can harm performance.

Within the same caption type, encoder performance varied by caption type. DistilBERT consistently outperformed MiniLM on text only captions across all backbones, with MAEs of 2.32, 1.88, and 1.92 against 2.40, 2.04, and 2.01 for DeiT, SwinV2, and MaxViT respectively. However, for hybrid captions the pattern reversed. MiniLM outperformed DistilBERT on all three backbones, with MAEs of 2.52 vs 2.59 on DeiT, 2.08 vs 2.36 on SwinV2, and 2.25 vs 2.28 on MaxViT. RemoteCLIP showed competitive performance on text only captions for DeiT, ranking second with an MAE of 2.38. However, for SwinV2 and MaxViT, MiniLM outperformed RemoteCLIP on text only captions with MAEs of 2.04 and 2.01 against 2.10 and 2.04 respectively. For hybrid captions, MiniLM performed most consistently, ranking first on DeiT (2.52) and SwinV2 (2.08). RemoteCLIP achieved the best hybrid result on MaxViT (2.14), while DistilBERT performed worst on hybrid captions for SwinV2 (2.36) and RemoteCLIP performed worst on DeiT (2.70), suggesting encoder suitability for hybrid captions varies by backbone capacity. Overall, the best performing configuration across all backbones was text only captions paired with DistilBERT.

C. Ablation Study: Text Scale

Following the identification of the best caption and encoder combination (text only captions paired with DistilBERT for all three backbones), a text scale ablation was conducted to analyze the sensitivity of multimodal fusion to the scaling factor α. Five values were evaluated for α, with results reported in Table 1.

TABLE I. TEXT SCALE ABLATION RESULTS

α	Deit ΔMAE	Deit %	Swin V2 ΔMAE	Swin V2 %	Max ViT ΔMAE	Max ViT %
0.3	0.27	9.51	0.55	22.01	0.23	9.71
0.5	0.49	17.31	0.58	23.53	0.39	16.76
0.7	0.55	19.54	0.61	24.36	0.39	16.77
1.0	0.54	19.29	0.54	21.70	0.32	13.57
1.5	0.63	22.26	0.51	20.65	0.44	18.81

Table 1 reports the test MAE improvement over the image only baseline (ΔMAE) and the corresponding relative percentage improvement for each backbone across all five α values. MAE is reported in proportion points on a 0–100 scale, where a value of 2.0 indicates an average prediction error of 2 percentage points per class. All three backbones show consistent improvement over the image only baseline across α values. However, the magnitude of improvement varies considerably with α, demonstrating that fusion strength is a meaningful design parameter. SwinV2 shows a non-monotonic response to α with a clear peak at α = 0.7, achieving its best performance with a 24.36% improvement and degrading at both extremes by dropping to 22.01% at α = 0.3 and 20.65% at α = 1.5. This suggests SwinV2 is the least sensitive to α in terms of absolute range. DeiT shows the largest sensitivity, improving monotonically from 9.51% at α = 0.3 to 22.26% at α = 1.5. MaxViT shows a mostly monotonic increase with α, with improvements ranging from 9.71% at α = 0.3 to 18.81% at α = 1.5, though gains are steady between α = 0.5 and α = 0.7. Overall, the results confirm that moderate to high fusion strength generalizes well across backbones, while very low α values consistently underutilize the available textual information.

D. Image Only vs. Best Multimodal Comparison

Table 2 presents the overall MAE comparison between the image only baseline and the best multimodal configuration per backbone. MAE is reported in proportion points (0–100 scale). Improvement percentages are computed as the relative reduction in MAE compared to the image only baseline. Across all three backbones, the best performing configuration selected based on validation MAE was text only captions generated by Qwen3-4B paired with DistilBERT, However, the optimal text scale differed per backbone. DeiT and MaxViT both achieved their best results at α = 1.5, while SwinV2 performed best at α = 0.7.

TABLE II. OVERALL MAE COMPARISON

Backbone	Image Only MAE	Best Multimodal MAE	Improvement (%)
DeiT	2.81	2.19	22.26
SwinV2	2.48	1.88	24.36
MaxViT	2.35	1.91	18.81

All three backbones benefit substantially from multimodal fusion. SwinV2 achieves the largest relative improvement of 24.36%, reducing MAE from 2.48 to 1.88. DeiT follows with a 22.26% improvement, reducing MAE from 2.81 to 2.19. MaxViT shows the smallest improvement of 18.81%, reducing MAE from 2.35 to 1.91. DeiT, the weakest backbone, achieves a comparable relative improvement to SwinV2 despite its lower capacity. These results are consistent with the hypothesis that higher capacity models extract richer visual representations, reducing their marginal gain from additional textual information.

E. Class-wise Performance

Table 3 presents the class-wise MAE for the best multimodal configuration per backbone, reported in proportion points.

TABLE III. CLASS-WISE MAE COMPARISON

Class	DeiT	SwinV2	MaxViT
Tree	2.59	2.57	2.39
Shrub	0.95	0.95	0.95
Grass	4.75	4.15	4.44
Crop	3.05	2.64	2.92
Built-up	0.65	0.43	0.47
Barren	2.52	2.02	1.80
Water	0.81	0.39	0.38

Grass consistently yields the highest MAE across all backbones, with scores of 4.75, 4.15, and 4.44 for DeiT, SwinV2, and MaxViT respectively indicating the class is hardest to predict. Crop follows as the second most challenging class with MAEs of 3.05, 2.64, and 2.92. Tree and Barren show moderate error levels, ranging from 2.39 to 2.59 for Tree and 1.80 to 2.52 for Barren across backbones. Shrub is remarkably consistent across all three backbones with an MAE of 0.95 in all cases. Built-up and Water also remain low across all configurations, with Built-up ranging from 0.43 to 0.65 and Water from 0.38 to 0.81. Comparing across backbones, no single architecture dominates all classes. SwinV2 achieves the best MAE on Grass (4.15), Crop (2.64), and Built-up (0.43), performing strongest on spatially dominant and high-coverage classes. MaxViT performs best on Tree (2.39), Barren (1.80), and Water (0.38), suggesting its stronger visual backbone better captures sparse and boundary-defined classes.

At the class level, multimodal fusion provides the most consistent benefit for Grass and Crop, which are the hardest classes overall. Comparing the best multimodal configuration against the image-only baseline per backbone, SwinV2 achieves the largest improvements on these classes with gains of 1.94 and 2.05 MAE points respectively, followed by DeiT with 1.62 and 1.98, and MaxVit with 1.30 and 1.21. Shrub shows near-zero improvement across all backbones. Water shows a small negative delta point for SwinV2 and near zero improvement for DeiT.

F. Error Analysis

Per-sample error distributions and qualitative prediction examples were examined across three best multimodal configurations. All backbones exhibit a right skewed error distribution. The qualitative inspection of the worst predicted samples reveals confusion between Shrub, Grass, and Tree due to overlapping spectral and textural features, and near-pure Barren scenes where the model distributes predicted proportions across vegetated classes instead.

Figure 1 shows a representative prediction example from MaxViT on a Crop dominant scene. As the regression head is unconstrained by design choice, predictions are clipped to zero for display purposes. The model correctly identifies Crop as the dominant class with a predicted proportion of 71.67 against a ground truth of 75.00, and estimates Grass (12.34 vs 11.00) and Barren (12.22 vs 14.00) accurately. Near zero classes such as Tree, Shrub, Built-up, and Water are predicted close to zero.

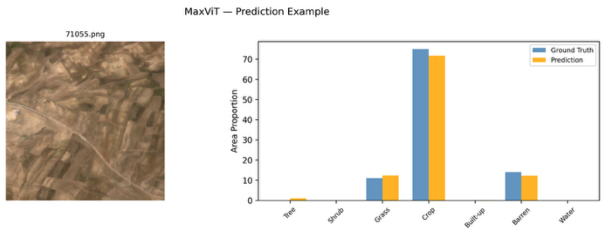

Fig. 1. MaxVit Prediction Example

V. Discussion

The results provide consistent evidence that textual descriptions improve land cover proportion estimation across all backbone architectures, directly confirming the primary research question. The magnitude of improvement, ranging from 18.81% to 24.36%, extends the findings of Tang et al. [3] to a continuous regression setting, and advances the regression-based land cover estimation approach of Sierra et al. [2] by showing that semantic captions further improve accuracy beyond visual features alone.

Caption type emerged as the most influential design choice. Factual text only captions consistently outperformed hybrid and vision based alternatives, directly reflecting the representation mismatch challenge identified by Xue and Zhu [1], as captions describing visual appearance rather than land cover composition fail to bridge the semantic gap to continuous regression targets. The weaker performance of vision based captions suggests that information extracted from the image and then converted back into text may be largely redundant with the visual representation already learned by the image encoder. This effect is more damaging for weaker backbones, with DeiT reaching an MAE of 3.10 under vision-based captions, suggesting lower capacity models have less ability to filter noisy inputs. The encoder ablation reveals that alignment between encoder capability and caption content matters more than pretraining domain. DistilBERT performs best on text only captions while MiniLM generalises more effectively to hybrid captions. RemoteCLIP's domain specific pretraining does not generalize consistently across backbone capacities and caption styles, a nuance not yet well explored in EO multimodal literature [4].

The text scale ablation confirms that fusion strength should be calibrated per backbone. Very small scaling factors consistently underutilized textual information, while moderate or high values generally produced better performance. SwinV2 peaks at $\alpha = 0.7$ with a clear optimum, DeiT improves monotonically up to $\alpha = 1.5$, and MaxViT shows diminishing returns at moderate scales. The comparison between image only and multimodal models provides insight into the relationship between backbone capacity and multimodal gains. MaxViT achieved the strongest image only baseline but exhibited the smallest relative improvement after incorporating text. Conversely, DeiT achieved a larger relative improvement despite its lower visual capacity. These findings support the hypothesis that weaker visual models rely more on complementary textual information, whereas stronger visual backbones are able to extract richer representations directly from imagery and therefore receive a smaller marginal benefit from text. In addition, no single backbone dominates across all classes. SwinV2 excels on high coverage classes while MaxViT leads on sparse boundary defined classes, likely reflecting architectural differences in attention mechanisms and suggesting ensemble approaches as a promising future direction. Multimodal fusion benefits Grass and Crop most, precisely the hardest classes, confirming that text is most valuable where visual ambiguity is greatest. The near zero Shrub improvement and small negative Water delta for SwinV2 suggest text adds little or can introduce noise for already well predicted classes.

Finally, the error analysis suggests that the remaining prediction errors are concentrated in a relatively small number of challenging samples. Most failures occur in scenes containing visually similar vegetation classes or unusual land cover compositions, indicating that dataset complexity rather than model instability is the primary source of residual error.

VI. Conclusion

This study investigated two research questions: whether textual descriptions of satellite images improve land cover proportion estimation compared to image only models, and whether backbone capacity moderates the benefit of multimodal fusion. Across three vision transformer backbones, three text encoders, three caption types, and five fusion strength values, multimodal fusion consistently improved performance with MAE reductions of 18.81% to 24.36%. Text only captions paired with DistilBERT proved most effective, and an inverse relationship between backbone capacity and multimodal gain was confirmed, as weaker backbones benefit more from textual information. Fusion strength should be calibrated per backbone, with weaker backbones benefiting from higher fusion weights. Grass and Crop benefited most from multimodal fusion while Shrub showed near zero improvement, suggesting text is most valuable where visual ambiguity is greatest. Persistent failure patterns on rare compositions point to dataset level limitations, suggesting future work should focus on targeted data augmentation and more flexible fusion architectures.

References

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